WEEK 5 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | Week Ending: 2nd October 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
Agent task contract	Completed	Goal, two tools, session state, limits and stop conditions documented. Registration pattern matches the code. A turn may use up to three tool calls so search, one rejected case call, and one correction fit. Students are treated as freshmen unless they say they are continuing, returning or postgraduate. The contract file is finished locally and will be committed under docs/requirements/.
Agent architecture diagram	Completed	Diagram on main shows the student chat, Sense–Plan–Act–Observe–Stop/Re-plan loop, Gemini requesting tools without executing them, the two tools, document store, draft store, session state, hard limits and stop conditions, with staff resolving drafts.
Working bounded agent workflow	Completed	Direct orchestration (not a framework). One student message runs the bounded loop: max 3 model rounds, max 3 tool calls per turn, create_support_case at most twice per turn, one draft per session. Agreement remains the sidebar control.
Three execution traces	Completed	T1: grounded webmail answer citing “Activating Your Makerere Webmail Account” (S1). T2: no_match then, after agreement, Wi-Fi draft to ICT helpdesk (S3 then S2). T3: rejects 23/A/93939, then saves draft with 23/U/22615 (failure then recovery).

Key Engineering Decisions

- Kept the same two Week 4 tools and enforced new Week 5 limits in the orchestration layer. The model may request a tool; the application decides whether it runs.

- Raised the per-turn tool budget to three calls so a search, a rejected create_support_case, and one corrected create_support_case can complete in one recovery path, matching the Agent Task Contract.
- Search stays at top-3 passages and similarity 0.70. A turn is labelled S1 only when the reply cites a source that retrieval actually returned.
- Students are assumed freshmen. Sense refuses only when the student says they are continuing, returning or postgraduate (S4). Free-text "yes" is never treated as case consent.
- If the model fails after a draft has already been saved, the student is still told the case id, and the failed call is recorded as an assistant turn so the next message is not a second user turn in a row.
- Architecture updates live on main. The bounded agent and traces remain on branch week5-execution-traces until merge.

Failures, Challenges and Current Response

- Gemini free-tier rate limit: the first failure-recovery run hit the five-requests-per-minute limit and saved no draft. Response: waited and re-ran; the saved T3 trace shows the bad registration number rejected, then the corrected number accepted.
- issue_summary still accepts a single character (Week 4 limitation carried forward). Response: left unchanged so the contract and traces stay consistent; tightening is deferred.
- Agent Task Contract is complete but not yet pushed to GitHub. Response: will be added under docs/requirements/ and linked from the repository.

Links and Evidence

GitHub repository: https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

Workflow and traces branch:

https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/Working_bounded_Agent_Workflow

https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/week5-execution-traces

ClickUp board (Team space, Week 5):
https://app.clickup.com/1200410000000374/v/b/li/1200410000022971

Individual Contribution Summary

Member	Task Owned	Evidence
Nakaye Hildah	Working bounded agent workflow	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/Working_bounded_Agent_Workflow
Namata Sumayya	Agent architecture diagram Week 5 progress report	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/Architecture/Group_M_eve_Agent%20Architecture%20Diagram.svg
Aryampa Joshua	Three execution traces	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/week5-execution-traces
Mushabe Moses	Agent task contract	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/requirements/Agent%20Task%20Contract%20M%20EVE.pdf

Plan for Next Week (Week 6 — Memory, State and Interoperability)

- Make workflow/session state explicit and retain only justified memory.
- Implement one justified persistent-memory use case (for example the draft already saved in this chat / active case), and document what is stored, why, who can access it, retention and deletion.
- Demonstrate that memory improves a legitimate task without silently controlling admissions, grading, disciplinary or fee decisions.
- Implement one external integration OR document one project capability as an MCP-style interface: capability, inputs, outputs, permissions and security boundary.
- Submit the Week 6 progress report with GitHub and ClickUp evidence.